PLamb Results Summary Report

Auto-generated from PLamb_results_summary.csv

Executive Summary

Total runs analyzed: 208
Unique primary models: 2
Unique DE parameterizations: 3
Dataset combinations observed: SN+BAO (85), SN (76), BAO (33), Unknown (14)

Lower AIC/BIC is better; lower χ^2/dof indicates a better per-dof fit.

Compare like-for-like dataset combos and link settings.

Leaderboards

Top by AIC (lower is better)

model	de_model	dataset_combo	AIC	AIC	BIC	bao_c_mode	c_model
nan	nan	Unknown	AIC 12.0 AIC 12.0 Name: 145, dtype: object	AIC 12.0 AIC 12.0 Name: 145, dtype: object	15.4	nan	nan
nan	nan	SN	AIC 12.391411 AIC 12.391411 Name: 110, dtype: object	AIC 12.391411 AIC 12.391411 Name: 110, dtype: object	12.121	nan	nan
nan	nan	BAO	AIC 12.4 AIC 12.4 Name: 153, dtype: object	AIC 12.4 AIC 12.4 Name: 153, dtype: object	12.9	nan	nan
nan	nan	Unknown	AIC 14.0 AIC 14.0 Name: 50, dtype: object	AIC 14.0 AIC 14.0 Name: 50, dtype: object	0	nan	nan
FR	lcdm	Unknown	AIC 14.0 AIC 14.0 Name: 79, dtype: object	AIC 14.0 AIC 14.0 Name: 79, dtype: object	0	c_of_z	linear_z
nan	nan	BAO	AIC 14.3 AIC 14.3 Name: 146, dtype: object	AIC 14.3 AIC 14.3 Name: 146, dtype: object	15.4	nan	nan
nan	nan	SN	AIC 22.166455 AIC 22.166455 Name: 29, dtype: object	AIC 22.166455 AIC 22.166455 Name: 29, dtype: object	28.261	nan	nan
nan	nan	BAO	AIC 23.8 AIC 23.8 Name: 151,	AIC 23.8 AIC 23.8 Name: 151,	28.3	nan	nan

			dtype: object	dtype: object			
nan	nan	BAO	AIC 23.8 AIC 23.8 Name: 150, dtype: object	AIC 23.8 AIC 23.8 Name: 150, dtype: object	28.4	nan	nan
nan	nan	BAO	AIC 23.8 AIC 23.8 Name: 149, dtype: object	AIC 23.8 AIC 23.8 Name: 149, dtype: object	28.4	nan	nan
FR	cpl	BAO	AIC 23.9 AIC 23.9 Name: 148, dtype: object	AIC 23.9 AIC 23.9 Name: 148, dtype: object	28.5	c_of_z	linear_z
nan	nan	BAO	AIC 24.1 AIC 24.1 Name: 152, dtype: object	AIC 24.1 AIC 24.1 Name: 152, dtype: object	28.6	nan	nan
nan	nan	BAO	AIC 24.1 AIC 24.1 Name: 147, dtype: object	AIC 24.1 AIC 24.1 Name: 147, dtype: object	28.6	nan	nan
nan	nan	BAO	AIC 25.663992 AIC 25.663992 Name: 154, dtype: object	AIC 25.663992 AIC 25.663992 Name: 154, dtype: object	25.556	nan	nan
nan	nan	BAO	AIC 40.115687 AIC 40.115687 Name: 133, dtype: object	AIC 40.115687 AIC 40.115687 Name: 133, dtype: object	40.008	nan	nan

Top by BIC (lower is better)

model	de_model	dataset_combo	BIC	AIC	BIC	bao_c_mode	c_model
FR	lcdm	Unknown	BIC 0.0	14	BIC 0.0	c_of_z	linear_z

			BIC 0.0 Name: 79, dtype: object		BIC 0.0 Name: 79, dtype: object		
nan	nan	Unknown	BIC 0.0 BIC 0.0 Name: 50, dtype: object	14	BIC 0.0 BIC 0.0 Name: 50, dtype: object	nan	nan
nan	nan	SN	BIC 12.12096 1 BIC 12.12096 1 Name: 110, dtype: object	12.391	BIC 12.120961 BIC 12.120961 Name: 110, dtype: object	nan	nan
nan	nan	BAO	BIC 12.9 BIC 12.9 Name: 153, dtype: object	12.4	BIC 12.9 BIC 12.9 Name: 153, dtype: object	nan	nan
nan	nan	Unknown	BIC 15.4 BIC 15.4 Name: 145, dtype: object	12	BIC 15.4 BIC 15.4 Name: 145, dtype: object	nan	nan
nan	nan	BAO	BIC 15.4 BIC 15.4 Name: 146, dtype: object	14.3	BIC 15.4 BIC 15.4 Name: 146, dtype: object	nan	nan
nan	nan	BAO	BIC 25.55581 2 BIC 25.55581 2 Name: 154, dtype: object	25.664	BIC 25.555812 BIC 25.555812 Name: 154, dtype: object	nan	nan
nan	nan	SN	BIC 28.26083 4 BIC 28.26083 4 Name: 29, dtype: object	22.166	BIC 28.260834 BIC 28.260834 Name: 29, dtype: object	nan	nan

nan	nan	BAO	BIC 28.3 BIC 28.3 Name: 151, dtype: object	23.8	BIC 28.3 BIC 28.3 Name: 151, dtype: object	nan	nan
nan	nan	BAO	BIC 28.4 BIC 28.4 Name: 150, dtype: object	23.8	BIC 28.4 BIC 28.4 Name: 150, dtype: object	nan	nan
nan	nan	BAO	BIC 28.4 BIC 28.4 Name: 149, dtype: object	23.8	BIC 28.4 BIC 28.4 Name: 149, dtype: object	nan	nan
FR	cpl	BAO	BIC 28.5 BIC 28.5 Name: 148, dtype: object	23.9	BIC 28.5 BIC 28.5 Name: 148, dtype: object	c_of_z	linear_z
nan	nan	BAO	BIC 28.6 BIC 28.6 Name: 147, dtype: object	24.1	BIC 28.6 BIC 28.6 Name: 147, dtype: object	nan	Nan
nan	nan	BAO	BIC 28.6 BIC 28.6 Name: 152, dtype: object	24.1	BIC 28.6 BIC 28.6 Name: 152, dtype: object	nan	nan
nan	nan	BAO	BIC 40.007508 BIC 40.007508 Name: 133, dtype: object	40.116	BIC 40.007508 BIC 40.007508 Name: 133, dtype: object	nan	nan

Top by χ^2/dof (lower is better)

No data available.

Grouped Performance (Model / DE-Model / Dataset Combo)

Group summary (first 20 rows)

('model', '')	**('de_model', '')**	**('dataset_combo', '')**	**('AIC', 'count')**	**('AIC', 'median')**	**('AIC', 'min')**	**('BIC', 'count')**	**('BIC', 'median')**	**('BIC', 'min')**

			)	)		)	)	
FR	cpl	BAO	1	23.9	23.9	1	28.5	28.5
FR	lcdm	SN+BAO	2	2039.9	2039.9	2	2072.6	2072.6
FR	lcdm	Unknown	1	14	14	1	0	0
FR	nan	SN+BAO	22	2022.4	2019.8	22	2055	2052.5
Nan	nan	BAO	32	64.437	12.4	32	65.715	12.9
Nan	nan	SN	76	1375.2	12.391	76	1396.2	12.121
Nan	nan	SN+BAO	61	2063.9	2014.3	61	2095.8	2047
Nan	nan	Unknown	9	1987.4	12	9	2009.1	0

Models, Parameters, and Relation to Standard Theories

ΛCDM: Baseline model with Ω_m and Ω_Λ; w = −1.
CPL (Chevallier–Polarski–Linder): w(z) = w0 + wa·(1 − a) = w0 + wa·z/(1+z).
FR: extends with A_acc, n_acc (accretion-driven acceleration), γ_c (variable-c coupling in BAO), ε_M (SN absolute-magnitude evolution). Optional soft-link: ε_M ≈ κ·γ_c with width σ.
BAO c-modes/c-models (e.g., c0 vs c_of_z, log1pz) set assumptions for c(z) in DH/DM predictions.

Interpreting Key Parameters

ε_M: SN magnitude evolution with redshift (can absorb systematics/acceleration signals).
γ_c: Variable-c coupling affecting BAO distances.
A_acc, n_acc: Amplitude and scaling of accretion-driven acceleration in FR.
w0, wa: DE EoS parameters in CPL; ΛCDM if w0=−1, wa=0.

Datasets and Effective Sizes

Max SN count detected in runs: not reported

Counts by dataset combo

dataset_combo	count
SN+BAO	85
SN	76
BAO	33
Unknown	14

Best-in-Class per Dataset Combo

SN

Best AIC

Model	de_model	dataset_combo	AIC	AIC	BIC	bao_c_mode	c_model
Nan	nan	SN	AIC 12.391411 AIC 12.391411 Name: 110, dtype: object	AIC 12.391411 AIC 12.391411 Name: 110, dtype:	12.121	nan	nan

				object			

Best BIC

Model	de_model	dataset_combo	BIC	AIC	BIC	bao_c_mode	c_model
Nan	nan	SN	BIC 12.120961 BIC 12.120961 Name: 110, dtype: object	12.391	BIC 12.120961 BIC 12.120961 Name: 110, dtype: object	nan	nan

Unknown

Best AIC

Model	de_model	dataset_combo	AIC	AIC	BIC	bao_c_mode	c_model
Nan	nan	Unknown	AIC 12.0 AIC 12.0 Name: 145, dtype: object	AIC 12.0 AIC 12.0 Name: 145, dtype: object	15.4	nan	nan

Best BIC

Model	de_model	dataset_combo	BIC	AIC	BIC	bao_c_mode	c_model
Nan	nan	Unknown	BIC 0.0 BIC 0.0 Name: 50, dtype: object	14	BIC 0.0 BIC 0.0 Name: 50, dtype: object	nan	nan

SN+BAO

Best AIC

Model	de_model	dataset_combo	AIC	AIC	BIC	bao_c_mode	c_model
Nan	nan	SN+BAO	AIC 2014.3 AIC 2014.3 Name: 183, dtype: object	AIC 2014.3 AIC 2014.3 Name: 183, dtype: object	2047	nan	nan

Best BIC

Model	de_model	dataset_combo	BIC	AIC	BIC	bao_c_mode	c_model
Nan	nan	SN+BAO	BIC 2047.0 BIC 2047.0 Name: 183, dtype: object	2014.3	BIC 2047.0 BIC 2047.0 Name: 183, dtype: object	nan	nan

BAO

Best AIC

Model	de_model	dataset_combo	AIC	AIC	BIC	bao_c_mode	c_model
Nan	nan	BAO	AIC 12.4 AIC 12.4 Name: 153, dtype: object	AIC 12.4 AIC 12.4 Name: 153, dtype: object	12.9	nan	nan

Best BIC

Model	de_model	dataset_combo	BIC	AIC	BIC	bao_c_mode	c_model
Nan	nan	BAO	BIC 12.9 BIC 12.9 Name: 153, dtype: object	12.4	BIC 12.9 BIC 12.9 Name: 153, dtype: object	nan	nan

Caveats & QC

Prefer well-mixed chains; when soft-link penalties are active, compare totals-with-penalties.

Use BIC to penalize complexity strongly; compare within identical dataset combos and link settings.